Ch 10.2: Multi-Layer Neural Nets

Lecture 30 - CMSE 381

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Announcements

Last time:

- Single Layer Neural Nets

This lecture:

- Multi-layer Neural Nets
- Application to MNIST

Announcements:

- Hw 8 Due Sunday 11/16
- HW 9 Due Sunday 11/23
- Exam 3 11/24
- Project Due 12/5

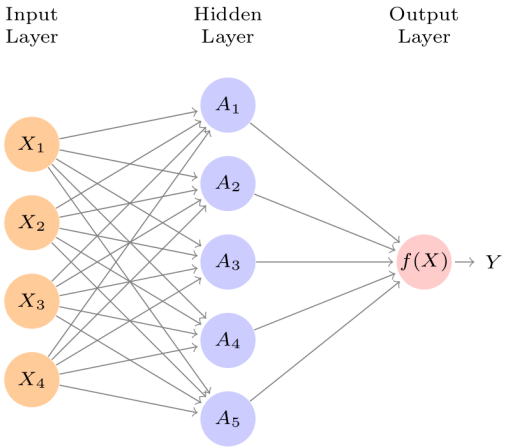
The end is near!

28	M	11/10	SVM	9.3, 9.4	Sun 11/9
29	W	11/12	Single Layer NN	10.1	
30	F	11/13	Multi Layer NN	10.2	HW #8 Due Sun 11/16
31	M	11/17	CNN	10.3	
32	W	11/19	Unsupervised learning / clustering	12.1, 12.4	
33	F	11/21	***Review***		HW #9 Due Sun 11/23
	M	11/24	**Midterm #3**		
	W	11/26	Virtual: Project Office Hours		
	F	11/28	Thanksgiving		
	M	12/1	Virtual: Project Office Hours		
	W	12/3	Virtual: Project Office Hours		
	F	12/5			**Project Due**
	M	12/8			
	W	12/10			
	F	12/12	**No final exam**		**Honors Project Due**

Section 1

Previously...

Feed Forward Neural Network: The cartoon



$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j),$$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

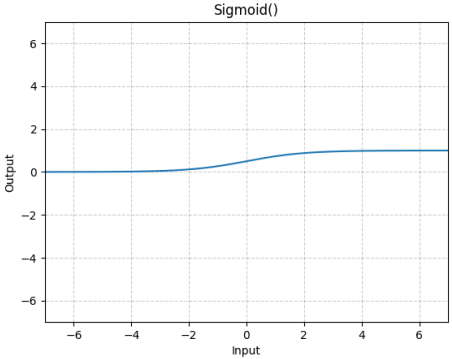
Test your understanding: PollEv

- Input layer is starting data
- Each arrow means taking combo of those values with weights to get next value
- Here there is one hidden layer then the output
- Get to pick how many hidden units K, here we have $K = 5$

Choices for activation function

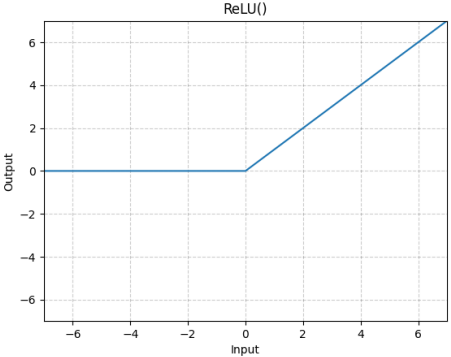
Sigmoid:

$$g(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

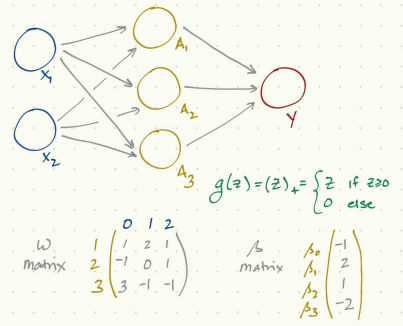


ReLU: Rectified linear unit

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$



Matrix version



$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$

$$A = g(\mathbf{W} \cdot \mathbf{X}) \qquad \mathbf{X}^T = (1\ X_1\ X_2\ \cdots\ X_p)$$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

$$Y = \beta \cdot \mathbf{A} \qquad \mathbf{A}^T = (1\ A_1\ A_2\ \cdots\ A_K)$$

Training the model

Choose parameters by minimizing RSS, $\sum_{i=1}^{n}(y_i - f(x_i))^2$ (or other loss function)

Chosen in advance:

- Number of layers (more on that in a moment)
- Number of hidden units
- Activation function $g(z)$

Taken together, these choices are called the "architecture"

Tuned by the model:

- ω
- β

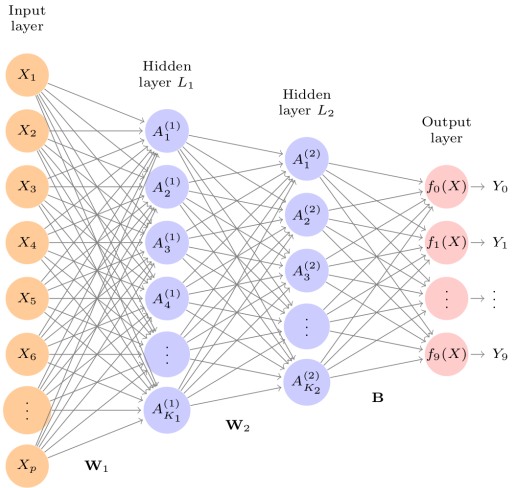
What will you learn today?

- What is the architecture of multilayer neural networks?
  - You should be able to describe what happens at each layer mathematically for a typical classification problem.
  - You should be able to calculate class probability for the output layer using softmax.
- How to classify images using multilayer neural networks in Python?

Section 2

Multilayer Neural Networks

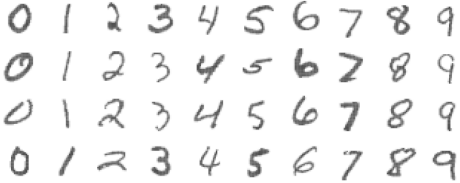
Multiple layers

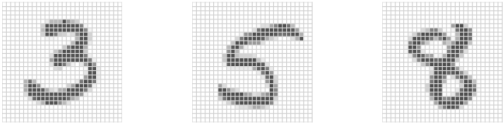


- Include more layers
- Can pick number of units per layer
- Each layer is linear combinations of previous, plus activation function

More typical for image classification

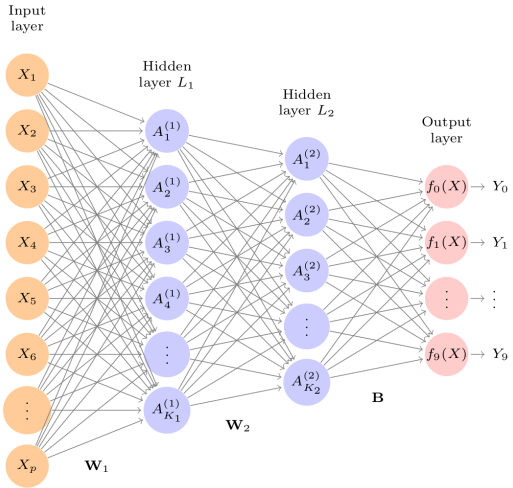
Example: MNIST





- Goal: Build a model to classify images into their correct digit class
- Each image has $p = 28 \cdot 28 = 784$ pixels
- Each pixel is grayscale value in [0,255]
- so we have 784 numbers as input, and 10 class as output

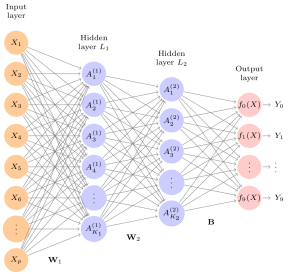
Hidden layers



$$
\begin{aligned}
A_k^{(1)} &= h_k^{(1)}(X) \\
&= g(w_{k0}^{(1)} + \textstyle\sum_{j=1}^{p} w_{kj}^{(1)} X_j) \\
A_\ell^{(2)} &= h_\ell^{(2)}(X) \\
&= g(w_{\ell 0}^{(2)} + \textstyle\sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)})
\end{aligned}
$$

- assuming we are classifying MNIST image (28x28=784 pixels)
- Two hidden layers. L_1 has 256 units; L_2 has 128
- 10 output variables due to class labeling

More on that architecture

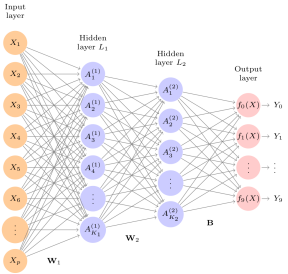


$$
\begin{aligned}
A_k^{(1)} &= h_k^{(1)}(X) \\
&= g(w_{k0}^{(1)} + \textstyle\sum_{j=1}^{p} w_{kj}^{(1)} X_j)
\end{aligned}
$$

$$
\begin{aligned}
A_\ell^{(2)} &= h_\ell^{(2)}(X) \\
&= g(w_{\ell 0}^{(2)} + \textstyle\sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)})
\end{aligned}
$$

- Superscript denotes layer
- W_1 denotes entire matrix of weight
- In this setting, size is $785 \times 256 = 200,960$ values
- 785 instead of 784 to involve intercept term (called bias in this literature)

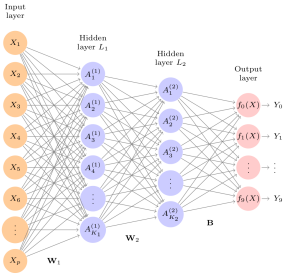
Matrix version: First layer



$$
\begin{aligned}
A_k^{(1)} &= h_k^{(1)}(X) \\
&= g(w_{k0}^{(1)} + \textstyle\sum_{j=1}^{p} w_{kj}^{(1)} X_j)
\end{aligned}
$$

$$A^{(1)} = g(\mathbf{W}^{(1)} \cdot \mathbf{X}) \qquad \mathbf{X}^T = (1\ X_1\ X_2\ \cdots\ X_p)$$

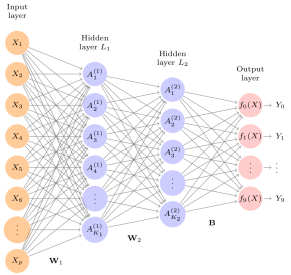
Matrix version: Second layer



$$
\begin{aligned}
A_\ell^{(2)} &= h_\ell^{(2)}(X) \\
&= g(w_{\ell 0}^{(2)} + \textstyle\sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)})
\end{aligned}
$$

$$
A^{(2)} = g(\mathbf{W}^{(2)} \cdot \mathbf{A}^{(1)}) \qquad (\mathbf{A}^{(1)})^T = (1 \; A_1^{(1)} \; A_2^{(1)} \; \cdots \; A_{K_1}^{(1)})
$$

Matrix version: Last layer, first step



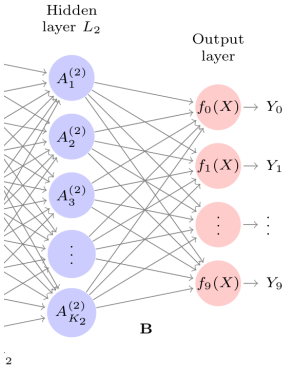
$$
\begin{aligned}
Z_m &= \beta_{m0} + \textstyle\sum_{\ell=1}^{K_2} \beta_{m\ell} h_\ell^{(2)}(X) \\
&= \beta_{m0} + \textstyle\sum_{\ell=1}^{K_2} \beta_{m\ell} A_\ell^{(2)},
\end{aligned}
$$

$$\mathbf{Z} = \beta \cdot \mathbf{A}^{(2)}$$

β is $M \times (K_2+1)$ matrix $\qquad (\mathbf{A}^{(2)})^T = (1\ A_1^{(2)}\ A_2^{(2)}\ \cdots\ A_{K_2}^{(2)})$

- Z is a temporary variable, later used in softmax function
- M is the number of classes

The last column for classification: Softmax



$$f_m(X) = \Pr(Y = m|X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},$$

- Want this to act like a probability.
- Answer is to use the softmax activation function $f_m(X)$
- Values are non-negative
- Values sum to 1
- Return class with highest probability

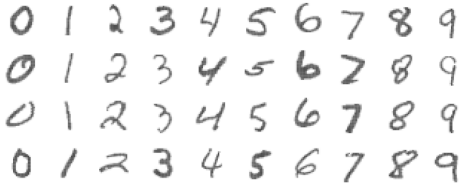
An example

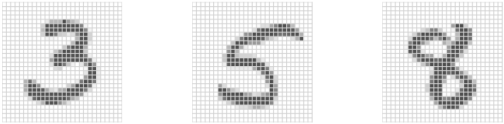
$$f_m(X) = \Pr(Y = m|X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},$$

$$Z = \begin{pmatrix} 1 & 3 & -1 & 2 & 5 \end{pmatrix}$$

- $e^1 + e^3 + e^{-1} + e^2 + e^5 = 178.97$
- $f = \begin{pmatrix} 0.01518 & 0.1122 & 0.00205 & 0.04128 & 0.82924 \end{pmatrix}$

MNIST



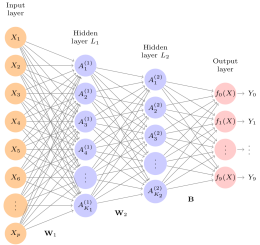


- Goal: Build a model to classify images into their correct digit class
- Each image has $p = 28 \cdot 28 = 784$ pixels
- Each pixel is grayscale value in [0,255]
- Data converted into column order
- Output represented by one-hot vector $Y = (Y_0, Y_1, \cdots, Y_9)$
- 60K training images, 10K test images

Coding

TL;DR

Feed Forward Neural Net



$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$

- Combines input data using learned weights
- Linear combo of those to get output
- Sometimes softmax to get probability of classification

Next time

21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/26
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	
24	F	10/31	Decision Trees	8.1	HW #6 Due Sun 11/2
25	M	11/3	Random Forests	8.2.1, 8.2.2	
26	W	11/5	Maximal Margin Classifier	9.1	
27	F	11/7	SVC	9.2	HW #7 Due Sun 11/9
28	M	11/10	SVM	9.3, 9.4	
29	W	11/12	Single Layer NN	10.1	
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	W	12/10			
	F	12/12	**No final exam**		**Honors Project Due**